Elasticity of interfaces: a multi-method approach

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1. Introduction

- Intentionally and controllably designed non-trivial material architectures can bring behaviour outperforming homogeneous bulk materials.
- Superlatttices represent perspective route for protective coatings. These, however, often require outstanding mechanical properties.
- For a knowledge-driven development of new systems, reliable and predictive modelling is required.
- *Ab initio* methods are accurate but computationally expensive; continuum mechanics treatment is affordable, but characterisation of interface and/or surface properties is required.

⇝ Method of Grimsditch and Nizzoli [1, 2]

2. The G&N method

Linear-elasticity continuum model-based on elasticity tensor, molar ratio and crystallographic orientation of each individual phase.

Limitations:

1. Neglects any heterogeneity introduced by the layer interface.
2. Neglects the bi-layer period.
3. The input C_{ij}s correspond to equilibrium lattice parameters of each phases, which can differ from the in-plane lattice parameter of the super-lattice.

G&N for 2 layers: $\hat{C}_{ij} = \chi(\hat{C}_1, f_1, \hat{C}_2, f_2)$

1. $\hat{P}_i = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$
2. $\hat{M}_i = \hat{P}_i^{-1}\hat{P}_1,\ i = 1, 2$
3. $\hat{M} = \sum_{i=1}^{2} f_i \hat{M}_i$
4. $C_{ij}(\text{bi-layer}) = \left(\sum_{i=1}^{2} f_i \hat{C}_i \hat{M}_i\right)\hat{M}^{-1}$

Can be generalised for N layers.

3. Conclusions

- An explicit *ab initio* elasticity calculations of nitride superlattices and a grain boundary in Ni_3Al intermetallic alloys.
- Continuum linear elasticity approach as formulated by Grimsditch and Nizzoli was applied.
- CrN/AlN superlattice yielded excellent agreement between explicit *ab initio* elastic tensor and its approximation by the G&N method.
- Qualitative agreement obtained also for majority of MoN/TaN systems.
- The G&N method fails for the case of Σ_5 GB in Ni_3Al.
- Reasons are:
  - omission of interface-induced modifications of atomic/electronic structure,
  - inability to reflect different bi-layer periods.

References

[1] M. Grimsditch, Phys. Rev. B Condens. Matter **31**, 6818 (1985).
[2] M. Grimsditch and F. Nizzoli, Phys. Rev. B Condens. Matter **33**, 5891 (1986)

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